

James Mann

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Education

University College London	Computer Science BSc	Sept 2024 – May 2027
• First Year Representative of the AI Society		
Carmel College	A Levels	Sept 2017 – May 2024
• A*A*A*AA (Maths, Further Maths, EPQ, Physics, Computer Science) • Elected Head Student		

Experience

Arcadia Impact	Research Assistant	May 2025 -
• Research Assistant responsible for identifying key stakeholders in the AI Evals ecosystem, interviewing them about their work, then aggregating the results into actionable technical directions with technical corroborations. • With an organisation contracted by the UK AI Security Institute to develop the open source Inspect Evals software used by Frontier Labs and leading AI Safety organisations.		
Impact Research Groups	Technical AI Safety Research Sprint	Feb 2025 - April 2025
• First author of a paper produced as part of an 8 week competitive research sprint, coming third. • Using mechanistic interpretability I identified a harmfulness direction, separate to refusal, that could steer between refusal and non-refusal, introducing the novel concept of a "tipping point" for refusal behaviour. • Proposed a theoretical framework for the success of jailbreaks as suppressors of harmfulness perception.		
Bruker	Machine Learning Experience Week	March 2023
• Integrated a Convolutional Neural Network with X-Ray Diffractometry, yielding a 100x speedup in deployment. • Developed a Streamlit application to interface between the data, the model, and the underlying technology.		

Projects / Hackathons

GPT from Scratch
• Created a transformer from scratch using NumPy with modular attention and multi-layer perceptron blocks. • With no additional dependencies designed an implementation of gradient descent and backpropagation to train the model for next token prediction.
Conditional Diffusion Model for Denoising Images
• Implemented a denoising diffusion probabilistic model in Pytorch, trained on CIFAR100 to generate novel images from the distribution, iteratively optimising the hyperparameters to improve the plausibility of samples. • Followed the paper Denoising Diffusion Probabilistic Models, implementing a UNet with attention heads
IdeaTracer at ICHack
• Developed IdeaTracer, a Claude powered agentic search engine aiming to build knowledge dependency graphs that trace the network of ideas over time.
DigiCapsule at UCLHack
• Aimed to solve the problem of trustless digital time capsules by creating unparallelisable computational proofs of work that guarantee access is after a set time.

Interests

- Rowing:** Compete for the university in the novice A boat.
- French:** Self teaching the language with an interest to study abroad.